

Improving Science Achievement—Is It Possible? Evaluating the Efficacy of a High School Chemistry and Physics Project-Based Learning Intervention

Barbara Schneider¹ , Joseph Krajcik¹, Jari Lavonen², Katariina Salmela-Aro²,
Christopher Klager¹, Lydia Bradford¹ , I-Chien Chen¹, Quinton Baker¹, Israel Touitou¹,
Deborah Peek-Brown¹, Rachel Marias Dezendorf¹, Sarah Maestrales¹, and Kayla Bartz¹

Crafting Engaging Science Environments is a high school chemistry and physics project-based learning intervention that meets *Next Generation Science Standards* performance expectations. It was administered to a diverse group of over 4,000 students in a randomized control trial in California and Michigan. Results show that treatment students, on average, performed 0.20 standard deviations higher than control students on an independently developed summative science assessment. Mediation analyses show an indirect path between teacher- and student-reported participation in modeling practices and science achievement. Exploratory analyses indicate positive treatment effects for enhancing college ambitions. Overall, results show that improving secondary school science learning is achievable with a coherent system comprising teacher and student learning experiences, professional learning, and formative unit assessments that support students in “doing” science.

Keywords: achievement; effect size; evaluation; experimental research; hierarchical linear modeling; learning environments; program evaluation; science education

U.S. students’ engagement and achievement in elementary and secondary science has been relatively stagnant over the past two decades (National Center for Education Statistics, 2017) and regrettably remains behind other industrialized countries (Organisation for Economic Co-operation and Development, 2020). Recognizing the consequences of persisting mediocre student academic performance in science achievement, three national efforts—*A Framework for K–12 Science Education* (National Research Council [NRC], 2012), the *Next Generation Science Standards* (NGSS; NGSS Lead States, 2013), and *Science and Engineering for Grades 6 to 12: Investigation and Design at the Center* (National Academies of Sciences, Engineering, and Medicine, 2019)—were initiated to articulate a vision and a set of standards for markedly reforming science teaching and learning. Breaking with traditional practices of memorizing science facts, formulae, and mathematical operations, these reports emphasize the importance of “doing” science for improving science literacy and encouraging the pursuit of science careers. Grounded in the ideas of John Dewey (1938)

and subsequent work by Blumenfeld et al. (1991) and Krajcik and Shin (2014), “the act of doing” engages the students in a cognitive process where they encounter a problem, plan a solution, work on it, and reflect on the results.

The NRC’s (2012) *Framework for K–12 Science Education* argues that science learning should focus on making sense of phenomena or solving design problems by engaging students in the three dimensions of scientific knowledge. These dimensions include science and engineering practices, crosscutting concepts, and disciplinary core ideas (DCIs) which should increase in-depth and sophistication within and across the grade levels (NRC, 2012). Science and engineering practices are behaviors scientists perform as they build theories about natural phenomena through investigations, creating models, generating explanations, and constructing science-based arguments. Crosscutting

¹Michigan State University, East Lansing, MI

²University of Helsinki, Helsinki, Finland

concepts refer to “ideas” that are found and linked across disciplines and which provide different tools for exploring phenomena. DCIs focus on the key ideas of a science discipline—or, in other instances, a fundamental organizing idea for a single discipline—and are critical for explaining phenomena.

Shortly after the development of the *Framework*, the NGSS were released having been created by a national effort of scientists, educational researchers, teachers, and policymakers (NGSS Lead States, 2013). The NGSS described a set of performance expectations that identified what students *should know and be able to do*. These performance expectations follow a progression of learning from primary to secondary grades, outlining and linking specific science ideas that increase in meaning and complexity. Many U.S. states have adopted the NGSS (National Science Teacher Association, 2017), or adapted them to develop similar science proficiency standards. Despite this widespread adoption of NGSS-like standards, there is a lack of research on evidence-based curricula and lessons that exemplify these reforms. The intervention “Crafting Engaging Science Environments” (CESE) was initiated to fill this gap by: (1) creating an intervention for high school chemistry and physics designed on the *Framework*’s vision and NGSS performance expectations and (2) assessing its effects on student science achievement (Schneider et al., 2020).

The Intervention: Crafting Engaging Science Environments

Deeply concerned with the lack of interest and engagement among secondary school students, an international team of Finnish and U.S. science education researchers, teachers, psychologists, sociologists, and psychometricians collaboratively designed the CESE intervention. Over the course of 3 years, this interdisciplinary team worked on developing a holistic intervention that takes a system approach to teaching and learning in secondary school chemistry and physics classrooms. The intervention consists of six units—three in chemistry and three in physics—with corresponding professional learning and assessments. Although team members from both countries collaborated on the design, the intent was not a comparative study but instead each country conducted its own investigations, recognizing the cultural and demographic differences of the student and teacher populations. Beginning with a design phase followed by a field-test conducted in the United States and Finland in 2017–2018 (Schneider et al., 2020), an efficacy trial was then conducted in the United States (2018–2019) with students of different demographic factors at the individual level (i.e., prior academic ability, race and ethnicity, and gender) and at the school level (i.e., urbanicity, percentage of free or reduced-price lunch, and region) in California and Michigan.¹ The results of this efficacy study are reported below.

Theoretical Basis of the Intervention

The CESE intervention was designed taking into account critiques of successful curricular interventions, including empirically tested instructional and learning materials for teachers and students, teacher professional learning with sustained support,

and three-dimensional assessments for evaluating learning in science (Harris et al., 2015; Polikoff et al., 2018; Sinatra et al., 2015). Guided by the principles of project-based learning (PBL) and the complementary ideas of the *Framework* and the NGSS, the coherent CESE system was formed to support science learning by challenging students to engage in relevant and meaningful experiences.

The theoretical principles of PBL in science revolve around learners making sense of “real-world phenomena” and solving relevant questions by planning and carrying out their own investigations (Krajcik & Shin, 2014). Collaborating with classmates, students engage in experiences in which they create artifacts that support the development of scientific ideas and use of scientific practices. PBL uses the three dimensions of scientific knowledge that allow students to draw from their knowledge across disciplines and life experiences, rather than being the passive recipients of knowledge. This contrasts with traditional approaches to chemistry and physics, where students are often instructed to plug numbers into equations without truly understanding the underlying relationships described by the equations.

Specifically, the CESE intervention focuses on seven major PBL principles: meeting subject and grade-relevant NGSS performance expectations; constructing a meaningful driving question that motivates a solution to a complex problem or an explanation for a compelling phenomenon; providing opportunities for the use of scientific practices; creating collaborative experiences and investigations for finding solutions to a driving question; integrating learning tools to make sense of evidence; developing artifacts that respond to the driving question and reveal students’ comprehension; and using assessments that capture emerging understandings (Krajcik & Shin, 2014). Enacting these principles helps transform the daily science experiences of teachers and students, from teacher-led instruction to environments where both teachers and students work together on solving problems and figuring out how to explain phenomena. In such a learning environment, students have agency and directly participate in using scientific practices, much like scientists and engineers.

There were several reasons for why our intervention design with its PBL framework was constructed for high school students in physics and chemistry. First, advances in science learning emphasize the components of PBL as necessary for student learning (National Academies of Sciences, Engineering, and Medicine, 2019; NRC, 2000). Second, there is limited work on measuring reforms in high school physical science courses (What Works Clearinghouse, 2020). We recognize that there are several newer studies that take a different approach to understanding why high school students lack an interest in science (National Assessment of Educational Progress, 2021) and why fewer students are choosing STEM (science, technology, engineering and mathematics) majors (Riegle-Crumb et al., 2011). Furthermore, there has also been many studies highlighting how different types of methodologies enhance students’ learning experiences in sciences (Lee et al., 2020).

There are considerable studies that examine student access among underrepresented minorities to more advanced level courses or subjects like chemistry or physics (National Science Board, 2020; Riegle-Crumb et al., 2019) with new promising

science curricular efforts (Engels et al., 2019; Sasson et al., 2018). However, several critiques concerned that the equitable and inclusionary participatory work of PBL suggest that it needs to be studied more rigorously to learn if indeed it has a positive impact on student outcomes (Chen & Yang, 2019; Cheung et al., 2016; Condliffe et al., 2017; Harris et al., 2019). Our work is a response to this critique.

Third, chemistry and physics are important subject areas, as they are often considered gatekeeper courses for many science specializations and postsecondary schooling (Hinojosa et al., 2016; Riegle-Crumb & King, 2010). Finally, scientific literacy is needed for all students, as evidenced by our understanding and responses when faced with pandemics, technological change, and environmental concerns (National Science Board, 2019). This combination of a need for new curricula paired with the necessity of PBL led to the creation of the CESE system.

Components of the Intervention

Teacher and Student Experiences and Materials

Recognizing the challenges that teachers would undoubtedly have in transforming instructional practices for all their science units, the team decided to develop three units for chemistry and physics, each of which lasted 4 to 6 weeks. Table 1 describes the three chemistry and three physics units, along with their driving questions, performance expectations, and phenomena.

Each unit was designed with an overriding driving question, lesson sequences incorporating scientific practices, and postunit assessments. The first step in the unit design process was to select a set of performance expectations for each unit in chemistry and physics and then unpack the performance expectations to elaborate on the ideas and identify the scientific practices (Harris et al., 2019; Krajcik & Czerniak, 2018; Krajcik & Shin, 2014).

The PBL framework requires each unit to have a driving question (see Table 1) that is meaningful to students' lives and connects a phenomena or complex problem to a concrete experience recognizable to the students. This drives the sequence of coherent lessons that continue to motivate students to meet the unit's learning goals. Moreover, the initial experiencing of the phenomena leads the students to ask their own questions.

Another important feature of the driving question is that it initiates the construction of a systematic sequence of lessons that builds throughout the unit, leading to additional questions that are threaded through various lessons, providing coherence across multiple experiences. The lessons that form from the driving question are not defined scripts but a flexible roadmap that connects prior experiences and ideas to specific scientific practices, such as planning and carrying out investigations, analyzing and interpreting data, and constructing explanations and designing solutions. The investigations related to the driving question are not independent projects, but rather carefully constructed experiential activities that build across the unit and are anchored in and help answer the driving question.

One of the most important scientific practices that the PBL units emphasize is having students construct models and connect them with evidence-based explanations of phenomena. Here again, the direct involvement of students in modeling is

not an isolated task, such as diagramming a simple relationship between two variables, but instead is directly related to explaining the phenomena under investigation and responding to the driving question. Raising the importance of modeling is a key scientific practice in the NRC's *Framework*, in which the intent is to provide students with experiences whereby they can become directly involved in systems thinking.²

By incorporating the practices of scientists and engineers, the modeling experiences afford students the opportunity to learn how to represent phenomena and physical systems, explain and predict the phenomena in a consistent and logical manner, and understand their data. The CESE modeling experiences are deliberately created so that students are supported in learning about identifying system components and the relationships among them which can take many forms, including mathematical formulae, diagrams, and computer simulations.

To understand how this process plays out in classrooms, the following two examples summarize the first units in chemistry and physics.³ The first unit in chemistry focuses on explaining evaporative cooling. Working from the CESE driving question (see Table 1), the lessons are designed so that students use classroom experiments and models to figure out and explain how evaporative cooling occurs; they must then figure out how this relates to the interactions of particles at the molecular level, as well as the matter's macro-level structure and properties, and energy transfer. Students manipulate different variables throughout the experiments, looking to explain how each component may influence evaporation. Across the unit, as students learn and assess how to make sense of phenomena, they construct models and explanations of the process of evaporative cooling, connecting energy changes to changes in the structure of matter in the system.

The first unit in physics focuses on students exploring the driving question, *How can I make a vehicle safer during a collision?* Here, experiences and investigations include working collaboratively to investigate and develop computer models to explain collisions, figuring out relationships among mass, force, and velocity in a collision by experiencing what happens when each of those variables is individually manipulated. Students use their new knowledge of mass, force, velocity, acceleration (Newton's second law), and momentum in combination with engineering practices to develop their best design in answering the driving question. Then they use a set of materials, to build and test a cart that minimizes the force on a passenger in a collision.

Postunit Assessments

One of the underlying principles of the CESE system is to create postunit assessment tasks that extend student learning experiences by using the three dimensions of scientific knowledge to explain phenomena and solve challenging problems to demonstrate mastery of NGSS performance expectations—but not what was articulated in the curricular unit. The steps used for creating these assessment tasks and rubrics were a modification of a previous process articulated by Harris et al. (2019). The development of assessment tasks allows for the creation of items through a principled, clearly defined process that is grounded in learning and assessment theory. All of the postunit assessments have the students draw models and write full descriptions of

Table 1
Units, Performance Expectations, Driving Questions and Phenomena for the Units

Unit	Performance expectation	Driving question	Phenomena
Evaporation	HS-PS1-3: Plan and conduct an investigation to gather evidence to compare the structure of substances at the bulk scale to infer the strength of electrical forces between particles. HP-PS3-2: Develop and use models to illustrate that energy at the macroscopic scale can be accounted for as a combination of energy associated with the motion of particles (objects) and energy associated with the relative positions of particles (objects).	“Why do I feel colder when I am wet than when I am dry?”	Water, acetone and ethanol evaporate when placed on your skin, making you feel cool. Ice at zero degrees Celsius will change to liquid water at zero degree with the addition of energy but no temperature change occurs.
Periodic table	HS-PS1-1: Use the periodic table as a model to predict the relative properties of elements based on the patterns of electrons in the outermost energy level of atoms. HS-PS1-2: Construct and revise an explanation for the outcome of a simple chemical reactions based on the outermost electron states of atoms, trends in the periodic table, and knowledge of the patterns of chemical properties.	“Why is table salt safe to eat, but the substances that form it are explosive or toxic when separated?”	Sodium reacts with water. Potassium reacts with water. A solution of sodium chlorine reacts with potassium iodide solution to form iodine and potassium chloride. A solution of bromine reacts with potassium iodide solution to form iodine and potassium bromide solution.
Conservation of matter	HS-PS1-7: Use mathematical representations to support the claim that atoms, and therefore mass, are conserved during a chemical reaction.	“Why does it seem like I can make a substance appear or disappear?”	Substances like paper, can burn. They appear to disappear, but in a closed system, the burning of paper has no mass change. It is necessary to add energy to start the burning of paper but after the start, lots of energy is given off as the temperature of the surrounding area increases.
Forces and motion	HS-PS2-1: Analyze data to support the claim that Newton’s second law of motion describes the mathematical relationship among the net force on a macroscopic object, its mass, and its acceleration. HS-PS2-3: Apply scientific and engineering ideas to design, evaluate, and refine a device that minimizes the force on a macroscopic object during a collision.	“How can I design a vehicle to be safer for a passenger during a collision?”	When a car crashes, destruction and damages to the car and the passengers occur.
Maglev	HS-PS3-5: Develop and use a model of two objects interacting through electric or magnetic fields to illustrate the forces between objects and the changes in energy of the objects due to the interaction; HS-PS3-2: Develop and use models to illustrate that energy at the macroscopic scale can be accounted for as a combination of energy associated with the motion of particles (objects) and energy associated with the relative positions of particles/ objects.	“What makes a super speed train (Maglev) function without touching the track?”	When a magnet is brought close enough to second magnet, the second magnet will move without touching it.
Electric motors	HS-PS3-1: Create a computational model to calculate the change in energy of one component in a system when the change in energy of the other components and energy flows in and out of the system are known; HS-PS2-5: Plan and conduct an investigation to provide evidence that an electric current can produce a magnetic field and that a changing magnetic field can produce an electric current; HS-PS3-3: Design, build, and refine a device that works within given constraints to convert one form of energy into another form of energy.	“How can I make the most efficient electric motor?”	An electric current can cause the shaft of an electric motor to spin or turn.

what is shown; these are then evaluated using a rubric that assesses their knowledge of the NGSS performance expectations, including the three dimensions of science learning. In addition, external reviews and classroom pilots were performed to increase item validity.

Professional Learning and Teacher Support

Professional learning was designed with best practices from research and emphasized teachers’ active participation in learning, connections to classroom contexts, collaboration, and

reflection (Darling-Hammond et al., 2017; Garet et al., 2001; Krajcik, 2014; van den Bergh et al., 2015). A key feature of CESE professional learning is for treatment teachers to experience what their students will be doing during their science activities, including using the driving question board, asking their own questions, building models, developing evidence-based explanations, and conducting experiments, not as students but as adult learners. The intent here is to guide and support the teachers in new ways of teaching often found to be challenging.

All treatment teachers spent three in-person days learning about the NGSS, three-dimensional learning, science PBL, and a review of the first units in chemistry and physics, conducted with team members and experienced teacher facilitators. Several times during the school year, the treatment teachers also met in person with facilitators to talk through teaching the next set of units. Facilitators also connected with the teachers via video conferences and online message boards. A hotline and a monitored email address were also available that centered on teacher questions. Over each of the 4- to 6-week units, there were approximately 90 teacher requests for additional support and information. If needed, facilitators were also available for face-to-face interactions.⁴ Control teachers also met in-person at the beginning of the school year for a day and were given a workshop on the NGSS.

Testing the Intervention

Three research questions guided this investigation:

Research Question 1: What is the main effect of this intervention on students' science learning? Do treatment students outperform control students on a summative science assessment?

Research Question 2: What other conditions besides the intervention could be affecting the treatment effect? More specifically, does the treatment effect vary by race/ethnicity and gender?

Research Question 3: What is the mediating effect of fidelity of implementation on the treatment effect?

Method

Sample

A power analysis indicated that the sample should include at least 48 schools with 50 students per school to detect a robust effect (i.e., 0.20).⁵ Schools were recruited from four areas (Los Angeles Unified School District, San Diego County Office of Education, Detroit Public School Community District, and other districts throughout Michigan), which allowed for a diverse sample of prior school-level science achievement, socioeconomic status, and race and ethnicity, including a substantial representation of Hispanic students, many of whom were English language learners. Participating school districts signed memoranda of understanding and supplied lists of schools for potential participation from which the randomization process was undertaken.

Randomization

Assignment to treatment status was made by schools rather than individual teachers to prevent spillover, as teachers within a particular school might plan together and/or share curriculum materials and instructional practices with colleagues. Also, a small number of schools requested that all their teachers in a subject be teaching the same curriculum. Team members contacted the principals from the district lists for potential participation, explaining that they would either be assigned to receive the treatment or control group that would receive the treatment the following school year. After receiving a principal's agreement to participate, schools were randomly assigned to treatment or control status, with an equal probability (0.50) of each. Given district and principal support, nearly all of the teachers were willing to participate. (See online Appendix A, available on the journal website, for the balance table between the treatment and control schools).⁶

Attrition

After randomization, nine schools attrited because of school closures or canceled science courses. Although not part of the sample selection (i.e., schools and students), teacher attrition was explored because of its impact on the student sample. Teacher attrition was related to school policies and personal issues: fiscal problems and a teacher strike which resulted in: teacher class reassignments and class cancellations; medical emergencies; mismatch between the course timeline and student abilities; and undisclosed reasons. Students were excluded from the analytic sample if they were missing either the pretest or summative assessment (or both).⁷ Students who joined classrooms after the pretest was administered were excluded from the analysis.

Initially, there were 70 schools (36 treatment and 34 controls), with 129 chemistry and physics teachers and 6,211 students (3,325 treatment and 2,886 controls). After accounting for attrition, the final analytic sample includes 61 schools (30 treatment and 31 controls), 119 teachers and 4,238 students (2,127 treatment and 2,111 controls). Table 2 summarizes overall and differential attrition across the treatment and control groups at the school and student levels.

Table 3 provides descriptive statistics and the balance for the analytic sample on pretest and demographic characteristics. The balance was estimated using a two-level hierarchical linear model (HLM) of the treatment on the characteristic of interest.

Given slight differences in the standardized pretest and proportion of race and ethnicity between the treatment and control students, these measures were included as control variables in the analytic models along with dummy variables for region and course subject.

Instruments

Students. Several instruments were used to collect information from the treatment and control students, including demographic characteristics, pretest baseline science achievement, and a summative assessment. For students whose first language was Spanish, all consent forms, teaching materials, and unit and summative

Table 2
Attrition

Level	Overall	Treatment	Control	Differential
Panel A: School level				
Initial schools	70	36	34	
Final schools	61	30	31	
Attrition	12.86%	16.67%	8.82%	7.84%
Panel B: Student level				
Initial students	6,211	3,325	2,886	
Final students	4,238	2,127	2,111	
Attrition	31.77%	36.03%	26.85%	9.18%

Table 3
Descriptive Statistics and Balance for the Analytic Sample

Characteristic	Overall		Control		Treatment		Difference (Treatment – Control)	Standardized mean difference
	Proportion	Number	Proportion	Number	Proportion	Number		
Standardized pretest	–0.08		–0.05		–0.11		–0.05* (0.03)	–0.06
Female	0.54	1047	0.54	1017	0.53		–0.00 (0.03)	–0.01
Missing female	0.09	157	0.07	225	0.11		0.00 (0.01)	0.01
White	0.29	587	0.28	629	0.30		0.06 (0.04)	0.12
Black	0.09	175	0.08	187	0.09		–0.04 (0.03)	–0.14
Hispanic	0.41	901	0.43	855	0.40		–0.01 (0.05)	–0.01
Asian	0.04	108	0.05	80	0.04		–0.01 (0.02)	–0.05
Other	0.02	42	0.02	35	0.02		–0.00 (0.01)	–0.02
Multiple races	0.03	78	0.04	67	0.03		–0.01 (0.01)	–0.03
Missing race	0.12	220	0.10	274	0.13		–0.00 (0.01)	–0.01
<i>N</i>	4,238	2,111		2,127				

Note. Standardized pretest is in standard deviations. Other covariates are proportions.
* $p < .05$. ** $p < .01$. *** $p < .001$.

tests were first translated into Spanish, and then translated back into English by another translator to ensure accuracy. All translators were fluent in both languages with a secondary school science specialization.

The student background survey was administered on Qualtrics or via paper and pencil. Questions were largely adapted from Programme for International Student Assessment (PISA) and included questions about home background (race, ethnicity, language, parent's education, etc.) attitudes toward science, and preferences for a future career in science. A pretest was given at the start of the school year to students in both conditions to measure their baseline science knowledge for a covariate in the analytic model. This pretest contained multiple choice items chosen from the National Assessment of Educational Progress

(NAEP) test bank. The items covered a range of topics and difficulty levels, plus, some were aligned with DCIs and performance expectations for chemistry and physics (see Maestrales, Zhai, et al., 2021).⁸

To measure the difference between the treatment and control groups at the end of the intervention, the students completed a summative assessment consisting of items developed by the Michigan Department of Education for use on the state's 11th-grade science assessment.⁹ Michigan was one of the earlier states to adopt NGSS-like standards and that adoption spurred an interest in a redesign of their science achievement test given to high school students. Several science curriculum specialists and psychometricians worked to design a science assessment that encompassed grade-level NGSS standards and NRC three-dimensional science.

Our summative test includes items that corresponded to the physical science performance expectations.¹⁰ This third-party assessment allowed for an object measure of the differences in achievement between the treatment and control group.¹¹

Student exit surveys asked about the frequency of different PBL activities in their classroom, which allowed for a comparison of student versus teacher perceptions of fidelity of implementation. The exit survey also asked students to reflect on the importance of science in their own lives, interest in the future study of science, experiences in their science classrooms, and science materials available at their schools.

Teachers. A teacher survey with questions adapted from the Teaching and Learning International Survey (TALIS) included items on years of teaching experience, teaching methods, and attitude toward teaching. This survey also included teacher knowledge and exposure to NGSS and PBL to ensure that the beginning of the intervention of such knowledge between the treatment and control teachers. All teachers and students were also asked to complete an exit survey at the end of the year. Items on the teacher exit survey asked about their use of PBL, questions on coverage of performance expectations, and the quality of classroom resources that affected the intervention lessons. These measures also allowed for the testing of the assumption that the control teachers were teaching business as usual. The exit survey items used to determine the units covered by the control teachers, their practices, and their textbooks and curriculum tools can be found in the online Appendix B (available on the journal website).

In addition to these exit surveys, in-person observations of randomly selected teachers (in both treatment and control) were also conducted and used for determining fidelity of implementation. Another important use of these observations was to obtain information and confirm that the control teachers were conducting their science classrooms with business as usual, and not using PBL practices or a similar type of curriculum that emphasized CESE principles or experiential activities. Although unable to conduct observations for all teachers, these observations provided important insights into the use of PBL in their classrooms, independent observer assessments of PBL use, and how these measures corresponded with teacher self-reports in the exit surveys.

Analysis

To assess the effect of the treatment on science achievement and to account for clustering that occurs as a result of assignment of schools to treatment, a two-level HLM was used (Bloom, 2005; Raudenbush, 1997; Raudenbush & Bryk, 2002), with students clustered within schools.

Model 1:

$$Y_{ij} = \gamma_{00} + \gamma_{01}Treatment_j + \mathbf{S}_j\delta_j + \mathbf{X}_{ij}\gamma_{i0} + r_{0j} + \epsilon_{ij}$$

where Y_{ij} is the standardized summative assessment test score with student i in school j ; γ_{00} is the mean outcome of the control group; γ_{01} is the difference between the treatment and control group; $\mathbf{S}_j$ are the school-level covariates, including school

pretest mean and region; and δ_j are the coefficients on those covariates. $\mathbf{X}_{ij}$ are the individual-level covariates, including pretest, course (chemistry or physics), and gender of the students. γ_{i0} are the coefficients on these individual-level covariates. Finally, r_{0k} is the school-level error term, and ϵ_{ij} is the student-level error term.

Because race, ethnicity, and gender data for every student was not available, Model 2 was estimated including a dummy variable for this missing data:

Model 2:

$$Y_{ij} = \gamma_{00} + \gamma_{01}Treatment_j + \mathbf{S}_j\delta_j + \mathbf{X}_{ij}\gamma_{i0} + \mathbf{M}_{ij}\alpha_{i0} + r_{0j} + \epsilon_{ij}$$

$\mathbf{X}_{ij}$ are the individual-level covariates, including pretest, course (chemistry or physics), gender, and now including race and ethnicity of the students. γ_{i0} are the coefficients on these individual-level covariates. $\mathbf{M}_{ij}$ is the vector of missing dummies for gender and race and α_{i0} are the coefficients on those dummies.

To determine whether treatment effects differ by race/ethnicity or gender, a cross-level interaction between gender and treatment and then race and treatment was conducted. The following model is specified for gender. The race and ethnicity model substitutes the Female interaction with the treatment with each Race dummy. The full model is shown in online Appendix C (available on the journal website).

Gender heterogeneity:

$$Y_{ij} = \gamma_{00} + \gamma_{01}Treatment_j + \gamma_{11}Treatment_j \times Female_{ij} + \mathbf{S}_j\delta_j + \mathbf{X}_{ij}\gamma_{i0} + \mathbf{M}_{ij}\alpha_{i0} + r_{0j} + \epsilon_{ij}$$

where γ_{11} is the coefficient on female treatment group.

When teachers implemented the intervention with fidelity, it was expected that students would have higher scores on their summative science assessment compared with students whose teachers did not. This assumption was based on teachers' reports of the frequency of various science practices, the extent to which they "incorporated" PBL into their classes (32% reported frequently, 50% sometimes, 15% rarely, 1% never), and students' reports of the use of modeling, aggregated to the teacher level to represent the extent to which teachers made this scientific practice a part of their science classroom instruction.

The impact of teacher fidelity of implementation was conducted with the 3-2-1 mediation model (Pituch et al., 2009). The 3-2-1 mediation model requires a three-level HLM where the teacher is at Level 2. This is because the expected mediation is from the teacher's level of implementing the intervention, which is a teacher-level variable. Therefore, to ensure that the treatment effects were consistent with the three levels, the overall treatment effect was estimated using a three-level HLM. Once the reliability of the three levels was found, the mediation analysis was conducted. Here, the treatment is delivered at Level 3 (the school level), and mediation occurs at Level 2 (the teacher level) and outcome is student assessment scores (Level 1). This process is conducted twice, first for the teachers' reported use of the intervention in their class and then the students' use of

Table 4
Estimated Treatment Effect of the CESE System

Effect	(1)	(2)
Treatment effect	0.220** (0.064)	0.208** (0.065)
Additional controls		×

Note. Treatment effect is the difference between the treatment and control group, measured in standard deviations. Standard errors are in parentheses. CESE = Crafting Engaging Science Environments.

* $p < .05$. ** $p < .01$. *** $p < .001$.

modeling in that teachers' class. The mediation model is shown in the online Appendix D (available on the journal website).

Additionally, one exploratory research question was whether the treatment would enhance education ambitions. For this model, a mixed effects logistic regression was used where the outcome was the change in levels of planned education attainment from the beginning to the end of the school year. The covariates used in this analysis were the same as Model 1 above. The models for estimating these effects are given in the online Appendix E (available on the journal website).

Generalizability of the Study

To assess the generalizability of the study, generalizability indexes were analyzed that summarized the degree of similarity between distributions of the propensity score of the sample schools compared with the various inference populations (the United States as a whole as well as each state; Tipton, 2014). This generalizability index takes values between 0 and 1. Essentially, for each of the 51 inference populations chosen (including the U.S. population as a whole), the sample of 61 schools was compared with the population of schools meeting the inclusion criteria (public schools) using a propensity score estimated with logistic regression. These scores estimate the probability that a school would be selected into the study given its covariates which included school enrollment, percentage of students on free or reduced-price lunch, school urbanicity, and percentage of White, Black, Hispanic, and Asian.

Results

Student Outcomes

Across both of the different models for the treatment effect, the treatment students outperformed the control students at a significance level of $p < .05$ or less and with a consistent treatment effect near 0.2 standard deviations. In the most conservative result as shown in Model 2, which includes all the covariates, the treatment students scored 0.208 standard deviations ($p < .01$) higher than the control students after taking into account the pretest, region, and demographic information. This represents about a 7 percentage-point increase for a standardized test score based on the summative assessment. Both the estimated treatment effects of the two models are reported in Table 4 (see online Appendix F, available on the journal website, for the full model results).

To confirm that the findings were robust, a sensitivity check was conducted using the Frank et al. (2013) framework for

evaluating the robustness of an inference. To invalidate this inference, 28.6% of the estimated treatment effect, which is approximately 1,637 observations, would have to be replaced with cases for which the effect of the treatment is zero.

Following the estimation of the treatment effect, Cohen's f^2 was calculated and found to be 0.01, indicating that the variance due to the treatment effect relative to the proportion of unexplained outcome variance is 0.01 (Lorah, 2018).

Tests of Heterogeneity for Research Question 2

In tests of heterogeneity by student demographic characteristics (gender and race/ethnicity), there were no statistically significant differences in treatment effect by student gender or race/ethnicity (see Table 5 and the online Appendix G (available on the journal website) for the full model of results).

As shown in Table 5, in the first iteration of the model, the outcome is significant for race/ethnicity and gender. However, when examining the effect for gender and race interacted with the treatment, the effects are not significantly different than zero. This interaction effect indicates that there is no evidence of a difference in the treatment by gender or race/ethnicity.

With regard to the exploratory model, a mixed effects logistic regression estimating change in educational ambitions from fall to spring was conducted. Students in the treatment group were 20% more likely than the control group to increase their postsecondary aspirations. The coefficient on the treatment was 0.18 (standard error = 0.09; $p = .05$). If treatment students originally intended to attend a 2-year school, they would be 1.2 times more likely at the end of the intervention to intend to attend a 4-year college than their control counterparts (see online Appendix H [available on the journal website] for the full model results).

Fidelity of Implementation

Table 6 shows the results of estimating the treatment effect using a three-level HLM. As seen in Table 6, the results are consistent with those above in the two-level HLM; therefore, it is appropriate to use a three-level model for the fidelity of implementation analysis.

Table 7 shows the results of the fidelity of implementation mediation models. The composite teacher measure of "incorporation of PBL" was not a statistically significant effect and only accounted for a small portion (11%) of the overall treatment effect (see Appendix I, available on the journal website, for the full model results).

However, students' reported use of modeling explains roughly 28% of the total treatment effect (and is significant at the .10 level). This was expected, as the teacher is ultimately responsible for guiding and supporting the students' modeling activities. The extent to which teachers' frequent use of modeling—one of the key scientific practices incorporated in the intervention—was a promising sign that this experience was a path through which the treatment worked.

Findings of Generalizability Index

When the generalizable index analysis was conducted using the Common Core Data with seven covariates, the results show that the generalizable index for the entirety of the United States is

Table 5
Summary of Student-Level Heterogeneity

Effect	Female	Black	Hispanic	Asian	Other race	Multiple races
Treatment	0.185** (0.708)	0.192** (0.069)	0.222** (0.085)	0.212** (0.064)	0.205** (0.065)	0.212** (0.066)
Predictor of interest	−0.061* (0.030)	−0.280** (0.087)	−0.238** (0.084)	0.044 (0.062)	−0.188 (0.096)	−0.086 (0.163)
Interaction	0.047 (0.058)	0.189 (0.120)	−0.033 (0.089)	−0.090 (0.139)	0.183 (0.151)	−0.109 (0.201)

Note. Coefficients are measured in standard deviations. Standard errors are in parentheses.
* $p < .05$ ** $p < .01$.

Table 6
Estimated Treatment Effect of the CESE System with Three Levels

Effect	(3)	(4)
Treatment effect	0.211*** (0.059)	0.196** (0.060)
Additional controls		×
Three levels (student, school, and teacher)	×	×

Note. Treatment effect is the difference between the treatment and control group, measured in standard deviations. Standard errors are in parentheses. CESE = Crafting Engaging Science Environments.
* $p < .05$. ** $p < .01$. *** $p < .001$.

0.82. This indicates that the sample from this study is similar to the inference population—here, the United States with regard to the covariates selected. In this case, when statistical adjustments are used to find an average treatment effect, these would be approximately unbiased for the inference population.

Discussion

Results of this randomized controlled trial and its generalizability are an important contribution to science teaching and learning; its significant effects on over 4,000 diverse students in two different states are especially heartening given the few science interventions that have been rigorously tested at the high school level and shown to be effective at improving science learning. Results show that the intervention was effective at raising students' science learning as measured by an independently developed summative assessment. There was no evidence that the intervention produced different effects based on students' gender, race, or ethnicity.

It is important to underscore what these results mean and what they do and do not conclude. Since there is no difference here among racial and ethnic groups on the effect of the treatment, this does not mean that the intervention is engaging history, culture, or race and ethnicity sufficiently. One of the key principles of CESE states that science must be personally meaningful and of interest to the students to engage them in science. This happens by having students ask questions about scientific phenomenon and having them relate these ideas to their own lives. These principles of CESE and their enactment are quite different from the current critique of science learning and curriculum (see Lee et al., 2020;

Pinkard et al., 2017; Rosebery et al., 2016). Additionally, CESE science experiences are deliberately designed for different inclusionary group activities. These activities are uniquely designed to involve multiple groups of students in problem solving, writing explanations, and becoming more informed about their world as well as learning from one another. Our results show that contrary to other interventions that often fail to positively affect all students, our theoretical framework and experiential learning opportunities should benefit all groups on average.

The most important takeaway is that science academic performance related to the NGSS can be improved with an intervention that is created and implemented as a coherent system approach: an approach that includes teacher and student learning experiences, teacher professional learning, and formative unit assessments that incorporate the three dimensions, including—but not limited to—modeling and writing explanations. The treatment provided teachers with multiple professional learning experiences on how to enact PBL that underscored the importance of “doing science.” Many of the teacher practices emphasized in the treatment included having the students take primary ownership for solving problems, figuring out phenomenon, engaging in scientific practices, and learning the meaning of science concepts across multiple DCIs.

Results shown in Table 6 indicate that “modeling” was an indirect pathway that affected students' science achievement scores. It is no surprise that students who frequently participated in modeling activities had an advantage on the summative assessment, as modeling is one of the eight scientific practices emphasized in the NGSS. However, modeling is not an experience that occurs independent of instructional opportunities. That the treatment students reported that modeling was a practice they used on multiple occasions suggests that a key principle of the intervention was being implemented in the classrooms.

Our goal has been to help adolescents not only become more scientific literate but also to ignite a new deepened interest in science, which in our exploratory work, we were pleased to find. Our exploratory analyses indicated that the intervention changed students' education ambitions. This was encouraging, as educational ambitions are a key predictor for college enrollment (Schneider et al., 2016). Given that the treatment increased students' college ambitions suggests that the intervention, with its emphasis on engaging in three dimensions of learning scientific knowledge to make sense of compelling phenomena or solve complex problems, may be a trigger for pursuing further education in science and other fields.

Table 7
Mediation Effects

Measure	a	b	Indirect effect (a * b)	95% Confidence interval	Total effect explained (a * b)/c
Teachers' incorporation of PBL	0.329 (0.126)	0.068 (0.055)	0.022 (0.021)	[-0.013, 0.071]	11%
Students' use of modeling	0.234 (0.056)	0.234 (0.139)	0.055 (0.036)	[-0.008, 0.132]	28%

Note. Standard errors are in parentheses. PBL = project-based learning.

Limitations of the Study

The three units in this intervention lasted, on average, 12 to 16 weeks. Most science courses typically last longer and include additional areas of study. Had the intervention lasted longer and included more units, the treatment effects on science learning may have been larger. However, it could also be the case that students may have reached a saturation point in their exposure to scientific practices and that teachers would be unable to sustain the types of instruction used in this intervention. From interviews we conducted with treatment teachers who participated in earlier phases of this study, this did not appear to be the case; indeed, teachers reported that they found themselves using these practices in other units that were not part of the CESE curriculum.

The lack of an effect for teacher reports in the meditation model may be the result of the measures employed. Other methods may have provided a more robust measure, such as many more in-person observations of teachers with high interrater agreement between multiple observers and repeated student surveys. However, due to cost constraints, this was not a possibility.

Finally, conducting a study of this magnitude—one that is a large-scale randomized control trial in two different states, and includes professional development and materials—is quite costly from financial, personnel considerations, and time-consuming. For future studies that look to expand and generalize this work to larger populations, these costs will be considerable. However, the promise of this intervention appears to be positive enough to warrant further investment.

Conclusion

Unquestionably, there is an immediacy for dramatically transforming science learning, especially given the health and environmental challenges young people are facing today and likely to face in the future. While recent major reforms to science teaching and learning have been met with enthusiasm and action at the state level, these reforms have not yet been widely adopted at the classroom level, particularly in high school chemistry and physics. This is in part because of the lack of aligned curriculum materials, professional learning, and assessments. In this respect, the results of this intervention are especially encouraging and merit further expansion.

This intervention is grounded in the recommendations of *A Framework for K–12 Science Education* (NRC, 2012) and *Science and Engineering for Grades 6 to 12* (National Academies of Sciences, Engineering, and Medicine, 2019), which incorporate

the three dimensions of scientific knowledge. Our results suggest it is possible to change the scientific learning environments for all students and expect positive science achievement results. However, this can only happen when there is a principled design system that involves not only just an engaging curriculum but also high-quality professional learning and formative assessments designed to stimulate knowledge. If science and science education community intend to improve science learning, students need to work on relevant meaningful problems and participate in scientific practices similar to the actual work of scientists—such as “figuring out” phenomena, building and testing models that explain those phenomena, searching for patterns and connections in data, and uncovering cause and effect relationships.

ORCID IDS

Barbara Schneider  <https://orcid.org/0000-0002-3233-166X>

Lydia Bradford  <https://orcid.org/0000-0003-2174-5565>

NOTES

This study is supported by the National Science Foundation (OISE-1545684; PIs Barbara Schneider and Joe Krajcik). Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation. Finnish authors were supported by the Academy of Finland (#298323; PIs Jari Lavonen and Katariina Salmela-Aro).

We thank the following people for their contributions and consultation regarding this study: Larry Hedges (Northwestern University), Jeffery Wooldridge (Michigan State University), Mark Reckase (Michigan State University), Yiling Cheng (Kaohsiung Medical University), and Kalle Juuti and Janna Inkinen (University of Helsinki).

¹The Finnish randomized controlled trial started a year later in 2019–2020 and had several interruptions because of the COVID-19 pandemic. Results from the Finnish efficacy study and U.S. control teachers who gave their new chemistry and physics classrooms the CESE treatment in 2019–2020 can be found in Maestres, Dezendorf, et al. (2021).

²See Cooper (2020) for a description of the relationship between crosscutting ideas, system thinking, and modeling.

³Additional details can be found on the CESE website: <https://sites.google.com/a/msu.edu/craftingengagingscience/home>

⁴On average for both chemistry and physics, respectively, the total professional learning was 40 hours.

⁵These estimates were based on *Optimal Design Software* (Spybrook et al., 2011), which calculates the number of clusters (i.e., schools) to power a study given the expected characteristics of the sample and a hypothesized effect size. This includes the number of observations (i.e., students), intraclass correlations, and R^2 values from previous

science assessments and covariates. These values were estimated based on information compiled by Spybrook et al. (2016), as well as our knowledge about the schools we expected to recruit.

⁶Because of space limitations, we have included an online Appendix for additional tables (available on the journal website) and information.

⁷If pretests were missing at random, imputation would not increase efficiency. The standard errors did not increase after imputation. If the imputation reduced the standard errors, then we would have expected greater efficiency of our models, but this was not the case (Wooldridge, 2010). The student attrition rate was higher than that recommended by the *What Works Clearinghouse* (2020), so as a robustness check we imputed pretest scores for those who only had a summative assessment, bringing our attrition rates down.

⁸To verify the quality of the pretest, a multinomial logistic regression and an item response theory (IRT) nominal response model were conducted to examine students' response patterns by gender (Cheng and Reckase [2020] for a fuller description of the pretest item differentiation). Results showed no significant differences on test-level scores between genders, or significant gender differences for most of the distractors. High-achieving girls and boys also chose the correct answers at the same frequency. These findings resonate with previous findings of gender similarities (Hyde & Linn, 2006; Zell et al., 2015).

⁹A confidentiality agreement with Michigan Department of Education was signed and the team was not allowed to show the items used for the summative assessment. Student scores and other de-identified information were stored on a secured server at Michigan State University.

¹⁰The Psychometric technical report for the science test has been delayed because of COVID-19. However, a test of the reliability of the items is available on request.

¹¹The physical science summative test scores for students taking chemistry or physics were made comparable using the R package *equateIRT*. The scores were then standardized for subjects and a 2pl model was analyzed. Then the two tests were equated with only the control students so that a treatment effect would not bias the equating process so that the students' summative assessments for physics and chemistry would be comparable based on the distribution of the two sets of scores (see online Appendix J [available on the journal website] for the full table of item equivalence for the chemistry and physics summative assessments).

REFERENCES

- Bloom, H. (2005). *Randomizing groups to evaluate place-based programs*. MDRC.
- Blumenfeld, P. C., Soloway, E., Marx, R. W., Krajcik, J. S., Guzdial, M., & Palincsar, A. (1991). Motivating project-based learning: sustaining the doing, supporting the learning. *Educational Psychologist*, 26(3–4), 369–398. <https://doi.org/10.1080/00461520.1991.9653139>
- Chen, C.-H., & Yang, Y.-C. (2019). Revisiting the effects of project-based learning on students' academic achievement: A meta-analysis investigating moderators. *Educational Research Review*, 26(February), 71–81. <https://doi.org/10.1016/j.edurev.2018.11.001>
- Cheng, Y., & Reckase, M. (2020). *The effect of gender differences and similarities on science performance* [Conference section]. American Educational Research Association Conference, San Francisco, CA, United States. <https://www.aera.net/Events-Meetings/Annual-Meeting/Previous-Annual-Meetings/2020-Annual-Meeting> (Conference canceled)
- Cheung, A., Slavin, R. E., Kim, E., & Lake, C. (2016). Effective secondary science programs: A best-evidence synthesis. *Journal of Research in Science Teaching*, 54(1), 58–81. <https://doi.org/10.1002/tea.21338>
- Condliffe, B., Quint, J., Visher, M. G., Bangser, M. R., Drohojowska, S., Saco, L., & Nelson, E. (2017). *Project-based learning: A literature review* (Working Paper). MDRC. <https://eric.ed.gov/?id=ED578933>
- Cooper, M. M. (2020). The crosscutting concepts: Critical component or “third wheel” of three-dimensional learning? *Journal of Chemical Education*, 97(4), 903–909. <https://doi.org/10.1021/acs.jchemed.9b01134>
- Darling-Hammond, L., Hyler, M. E., & Gardner, M. (2017, May). *Effective teacher professional development* [Research Brief]. Learning Policy Institute.
- Dewey, J. (1938). *Experience & education*. Kappa Delta Pi.
- Engels, M., Miller, B., Squires, A., Jennewein, J. S., & Eitel, K. (2019). The confluence approach: Developing scientific literacy through project-based learning and place-based education in the context of NGSS. *Electronic Journal of Science Education*, 23(3), 33–58. <https://files.eric.ed.gov/fulltext/EJ1228452.pdf>
- Frank, K. A., Maroulis, S. J., Duong, M. Q., & Kelcey, B. M. (2013). What would it take to change an inference? Using Rubin's causal model to interpret the robustness of causal inferences. *Educational Evaluation and Policy Analysis*, 35(4), 437–460. <https://doi.org/10.3102/0162373713493129>
- Garet, M. S., Porter, A. C., Desimone, L., Birman, B. F., & Yoon, K. S. (2001). What makes professional development effective? Results from a national sample of teachers. *American Educational Research Journal*, 38(4), 915–945. <https://doi.org/10.3102/00028312038004915>
- Harris, C. J., Krajcik, J. S., Pellegrino, J. W., & DeBarger, A. H. (2019). Designing knowledge-in-use assessments to promote deeper learning. *Educational Measurement: Issues and Practice*, 38(2), 53–67. <https://doi.org/10.1111/emip.12253>
- Harris, C. J., Penuel, W., D'Angelo, C., DeBarger, A., Gallagher, L., Kennedy, C., Cheng, B., & Krajcik, J. (2015). Impact of project-based curriculum materials on student learning in science: Results of a randomized controlled trial. *Journal of Research in Science Teaching*, 52(10), 1362–1385. <https://doi.org/10.1002/tea.21263>
- Hinojosa, T., Rapaport, A., Jaciw, A., LiCalsi, C., & Zacamy, J. (2016). *Exploring the foundations of the future STEM workforce: K–12 indicators of postsecondary STEM success*. U.S. Department of Education.
- Hyde, J. S., & Linn, M. C. (2006). Gender similarities in mathematics and science. *Science*, 314(5799), 599–600. <https://doi.org/10.1126/science.1132154>
- Krajcik, J. (2014). The importance of viable models in the construction of professional development. In B. S. Wojnowski & C. P. Pea (Eds.), *Models and approaches to STEM professional development* (pp. 33–44). NSTA Press. <https://static.nsta.org/pdfs/samples/PB322Xf.pdf>
- Krajcik, J., & Czerniak, C. (2018). *Teaching science in elementary and middle school: A project-based approach* (5th ed.). Taylor & Francis.
- Krajcik, J., & Shin, N. (2014). Project-based learning. In R. K. Sawyer (Ed.), *The Cambridge handbook of the learning sciences* (pp. 275–297). Cambridge University Press.
- Lee, C., Nasir, N., Pea, R., & de Royston, M. (2020). Reconceptualizing learning: A critical task for knowledge-building and teaching. In N. S. Nasir, C. D. Lee, R. Pea, & M. M. de Royston (Eds.), *Handbook of the cultural foundations of learning* (pp. xvii–xxv). Routledge.
- Lorah, J. (2018). Effect size measures for multilevel models: Definition, interpretation, and TIMSS example. *Large-scale Assessments in Education*, 6(1), Article 8. <https://doi.org/10.1186/s40536-018-0061-2>
- Maestraldes, S., Dezendorf, R., Tang, X., Salmela-Aro, K., Bartz, K., Juuti, K., Lavenon, J., Krajcik, J., & Schneider, B. (2021). U.S. and Finnish high school science engagement during the COVID-19 pandemic. *International Journal of Psychology*. Advance online publication. <https://doi.org/10.1002/ijop.12784>

- Maestres, S., Zhai, X., Toutou, I., Baker, Q., Schneider, B., & Krajcik, J. (2021). Using machine learning to evaluate multi-dimensional assessments of chemistry and physics. *Journal of Science Education and Technology*, 30(2), 239–254. <https://doi.org/10.1007/s10956-020-09895-9>
- National Academies of Sciences, Engineering, and Medicine. (2019). *Science and engineering for Grades 6–12: Investigation and design at the center*. The National Academies Press.
- National Assessment of Educational Progress. (2021). *Results from the 2019 Science Assessment*. U.S. Department of Education and the Institute of Education Sciences.
- National Center for Education Statistics. (2017). *The condition of education 2017*. Author. <https://nces.ed.gov/pubsearch/pubsinfo.asp?pubid=2017144>
- National Research Council. (2000). *How people learn: Brain, mind, experience, and school* (Expanded ed.). National Academies Press.
- National Research Council. (2012). *A framework for K–12 science education: Practices, crosscutting concepts, and core ideas*. National Academies Press.
- National Science Board. (2019). *The skilled technical workforce: Crafting America's science and engineering enterprise*. <https://www.nsf.gov/nsb/publications/2019/nsb201923.pdf>
- National Science Board. (2020). *The state of U.S. science and engineering 2020*. <https://nces.nsf.gov/pubs/nsb20201>
- National Science Teacher Association. (2017). *NGSS hub*. <https://ngss.nsta.org/>
- NGSS Lead States. (2013). *Next generation science standards: For states, by states*. National Academies Press. https://epsc.wustl.edu/seismology/book/presentations/2014_Promotion/NGSS_2013.pdf
- Organisation for Economic Co-operation and Development. (2020). *Science performance (PISA)—indicator*. <https://data.oecd.org/pisa/science-performance-pisa.htm>
- Pinkard, N., Erete, S., Martin, C., & McKinney de Royston, M. (2017). Digital Youth Divas: Exploring narrative-driven curriculum to spark middle school girls' interest in computational activities. *Journal of the Learning Sciences*, 26(3), 477–516. <https://doi.org/10.1080/10508406.2017.1307199>
- Pituch, K. A., Murphy, D. L., & Tate, R. L. (2009). Three-level models for indirect effects in school- and class-randomized experiments in education. *Journal of Experimental Education*, 78(1), 60–95. <https://doi.org/10.1080/00220970903224685>
- Polikoff, M. S., Campbell, S. E., Koedel, C., Le, Q. T., Haraway, T., & Gasparian, H. (2018). *The formalized processes districts use to evaluate textbooks* [University of Southern California Working Paper].
- Raudenbush, S. W. (1997). Statistical analysis and optimal design for cluster randomized trials. *Psychological Methods*, 2(2), 173–185. <https://doi.org/10.1037/1082-989X.2.2.173>
- Raudenbush, S. W., & Bryk, A. (2002). *Hierarchical linear models* (2nd ed.). Sage.
- Riegle-Crumb, C., & King, B. (2010). Questioning a White male advantage in STEM: Examining disparities in college major by gender and race/ethnicity. *Educational Researcher*, 39(9), 656–664. <https://doi.org/10.3102/0013189X10391657>
- Riegle-Crumb, C., King, B., & Irizarry, Y. (2019). Does STEM stand out? Examining racial/ethnic gaps in persistence across postsecondary fields. *Educational Researcher*, 48(3), 133–144. <https://doi.org/10.3102/0013189X19831006>
- Riegle-Crumb, C., Moore, C., & Ramos-Wada, A. (2011). Who wants to have a career in science or math? Exploring adolescents' future aspirations by gender and race/ethnicity. *Science Education*, 95(3), 458–476. <https://doi.org/10.1002/sce.20431>
- Rosebery, A. S., Warren, B., & Tucker-Raymond, E. (2016). Developing interpretive power in science teaching. *Journal of Research in Science Teaching*, 53(10), 1571–1600. <https://doi.org/10.1002/tea.21267>
- Sasson, I., Yehuda, I., & Malkinson, N. (2018). Fostering the skills of critical thinking and question-posing in a project-based learning environment. *Thinking Skills and Creativity*, 29(September), 203–212. <https://doi.org/10.1016/j.TSC.2018.08.001>
- Schneider, B., Klager, C., Chen, I.-C., & Burns, J. (2016). Transitioning into adulthood: Striking a balance between support and independence. *Policy Insights From the Behavioral and Brain Sciences*, 3(1), 106–113. <https://doi.org/10.1177/2372732215624932>
- Schneider, B., Krajcik, J., Lavonen, J., & Salmela-Aro, K. (2020). *Learning science: The value of crafting engagement in science environments*. Yale University Press. <https://doi.org/10.2307/j.ctvwjckf1>
- Sinatra, G. M., Heddy, B. C., & Lombardi, D. (2015). The challenges of defining and measuring student engagement in science. *Educational Psychologist*, 50(1), 1–13. <https://doi.org/10.1080/00461520.2014.1002924>
- Spybrook, J., Bloom, H., Congdon, R., Hill, C., Martinez, A., & Raudenbush, S. W. (2011). *Optimal design plus empirical evidence: Documentation for the "optimal design."* <http://www.hlmsft.net/od/od-manual-20111016-v300.pdf>
- Spybrook, J., Westine, C. D., & Taylor, J. A. (2016). Design parameters for impact research in science education: A multistate analysis. *AERA Open*, 2(1). <https://doi.org/10.1177/2332858415625975>
- Tipton, E. (2014). How generalizable is your experiment? An index for comparing experimental samples and populations. *Journal of Educational and Behavioral Statistics*, 39(6), 478–501. <https://doi.org/10.3102/1076998614558486>
- van den Bergh, L., Ros, A., & Beijard, D. (2015). Teacher learning in the context of a continuing professional development programme: A case study. *Teaching and Teacher Education*, 47(April), 142–150. <https://doi.org/10.1016/j.tate.2015.01.002>
- What Works Clearinghouse. (2020). *What Works Clearinghouse™: Standards handbook, version 4.1*. Institute of Education Sciences, U.S. Department of Education. <https://ies.ed.gov/ncee/wwc/Docs/referenceresources/WWC-Standards-Handbook-v4-1-508.pdf>
- Wooldridge, J. M. (2010). *Econometric analysis of cross section and panel data*. MIT Press.
- Zell, E., Krizan, Z., & Teeter, S. R. (2015). Evaluating gender similarities and differences using metasynthesis. *American Psychologist*, 70(1), 10–20. <https://doi.org/10.1037/a0038208>

AUTHORS

BARBARA SCHNEIDER, PhD, is the John A. Hannah University Distinguished Professor at Michigan State University College of Education, 516B Erickson Hall, 620 Farm Lane, East Lansing, MI 48823; bschneid@msu.edu. Her research focuses on how the social context of schools and families influence the academic and social well-being of adolescents as they move into adulthood.

JOSEPH KRAJCIK, PhD, is the Lappan-Phillips Professor at Michigan State University College of Education, 115 Erickson Hall, 620 Farm Lane, East Lansing, MI, 48823; krajcik@msu.edu. His research focuses on improving the teaching and learning of science, kindergarten through college by engaging in innovation and research.

JARI LAVONEN, PhD, is a professor of physics and chemistry education at the University of Helsinki, PL 9, Siltavuorenpenger 5A, 00014, Helsinki, Finland; jari.lavonen@helsinki.fi. His research focuses on science and technology education and teacher education.

KATARIINA SALMELA-ARO, PhD, is an academy professor in educational sciences at the University of Helsinki, PL 9, Siltavuorenpenger

5A, 00014, Helsinki, Finland; *katariina.salmela-aro@helsinki.fi*. Her research focuses on social psychology in teacher burnout, task-values, and motivation.

CHRISTOPHER KLAGER, PhD, is a research associate at the STEPP Center in the Institute for Policy Research at Northwestern University, 2040 Sheridan Rd, Evanston, IL 60208; *crklager@gmail.com*. His research focuses on the translation and communication of evidence about educational programs for policymakers and practitioners.

LYDIA BRADFORD is a graduate research assistant at Michigan State University College of Education, 516 Erickson Hall, 620 Farm Lane, East Lansing, MI 48823; *bradf134@msu.edu*. Her research focuses on statistical modeling, analysis, research design, and causal inference within the context of educational research.

I-CHIEN CHEN, PhD, is a research associate in the College of Education at Michigan State University, 516 Erickson Hall, 620 Farm Lane, East Lansing, MI 48823; *ichiench@msu.edu*. Her research focuses on how social contexts enhance social emotional learning, interests, and achievement.

QUINTON BAKER, PhD, is a research agricultural economist in the food markets branch of the Food Economic Division of the USDA, 805 Pennsylvania Avenue, Kansas City, MO 64105; *quinton.baker@usda.gov*. His research focuses on public policy, U.S. food assistance programs, labor economics, and food marketing.

ISRAEL TOUITOU, PhD, is a research associate at the CREATE for STEM Institute at Michigan State University, 620 Farm Lane, East

Lansing, MI 48823; *touitou@msu.edu*. His research focuses on project-based learning in secondary physics education and assessments.

DEBORAH PEEK-BROWN was a curriculum director at the college of education at Michigan State University, 620 Farm Lane, East Lansing, MI, 48823; *dpeekbrown@gmail.com*. She developed project-based science curriculum and professional learning.

RACHEL MARIAS DEZENDORF, PhD, is an analyst at the Academic Senate at the University of California, Berkeley, 320 Stephens Hall, Berkeley, CA 94720; *rmariasdezendorf@berkeley.edu*. She focuses on consulting and coordinating courses at the university.

SARAH MAESTRALES is a graduate research assistant at Michigan State University College of Education, 516 Erickson Hall, 620 Farm Lane, East Lansing, MI 48823; *maestral@msu.edu*. Her research focuses on large-scale data analysis and survey development.

KAYLA BARTZ is a graduate research assistant at Michigan State University College of Education, 516 Erickson Hall, 620 Farm Lane, East Lansing, MI 48823; *bartzkay@msu.edu*. Her research focuses on test development and psychometric properties of varying assessments and surveys.

Manuscript received May 20, 2020
Revisions received February 5, 2021,
and September 21, 2021
Accepted October 29, 2021